Proof

Given : In $\Delta ABC, \frac{AD}{DB} = \frac{AE}{EC}$

To prove : $DE \parallel BC$

Construction : If DE is not parallel to BC, draw $DF \parallel BC$.

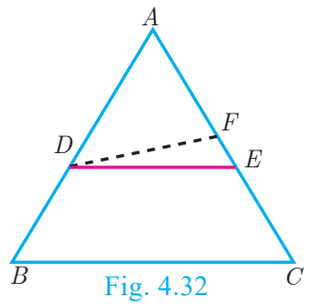


Fig. 4.32

No.	Statement	Reason
1.	$\frac{AD}{DB} = \frac{AE}{EC}$ ...(1)	Given
2.	$\Delta ABC, DF \parallel BC$	Construction
3.	$\frac{AD}{DB} = \frac{AF}{FC}$ ... (2)	Thales theorem
4.	$\frac{AE}{EC} = \frac{AF}{FC}$	From (1) and (2)
	$\frac{AE}{EC} + 1 = \frac{AF}{FC} + 1$	Adding 1 to both sides
	$\frac{AE + EC}{EC} = \frac{AF + FC}{FC}$	
	$\frac{AC}{EC} = \frac{AC}{FC}$	
	EC = FC	Cancelling AC on both sides
	Therefore, $E = F$	Our assumption that DE is not parallel to BC is wrong.
	Thus $DE \parallel BC$	Hence proved

Theorem 3: Angle Bisector Theorem

Statement

The internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the corresponding sides containing the angle.

Proof

Given : In Δ ABC, AD is the internal bisector

To prove : $\frac{AB}{AC} = \frac{BD}{CD}$

Construction : Draw a line through C parallel to AB. Extend AD to meet line through C at E

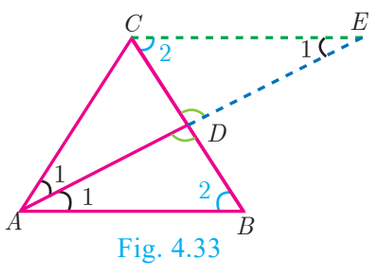


Fig. 4.33

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